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جَامِعَةُ الْقَاهِرَةِ
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Software Testing: Assignment 1

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“java-math-library” Testing Report

de.tilman_neumann.util.*StringUtil* Class Testing

A string utility Class

1. STRING REPEAT(STRING S, INT N)

Input Domain Modeling

Characteristics	Blocks		
A: String. Null	True	False	
B: String. Length	Empty	Equal to 1	Greater than 1
C: Number	<= 0 (boundary value)	Equal to 1	Greater than 1

Constraints: If the string is null -> length characteristic is not applicable.

Test Cases

The test coverage criteria chosen: **Multiple Base Choice Coverage (MBCC)**

Base test cases: **A2, B2, C2 & A2, B3, C3**

Tests	Blocks Covered	Status
T1	A2, B2, C2	Pass
T2	A2, B1, C2	Pass
T3	A2, B3, C2	Pass
T4	A1, C2	Pass
T5	A2, B2, C1	Pass
T6	A2, B2, C3	Pass
T7	A2, B3, C3	Pass
T8	A1, C3	Pass
T9	A2, B1, C3	Pass
T10	A2, B3, C1	Pass

2. STRING FORMATLEFT (STRING S, STRING MASK)

Input Domain Modeling

Characteristics	Blocks		
A: String. Null	True	False	
B: String. Length	Empty	Equal to 1	Greater than 1
C: Mask. Null	True	False	
D: Mask. Length	Empty	Equal to 1	Greater than 1

Constraints: If the string or the mask is **null** -> length characteristics are not applicable.

Test Cases

The test coverage criteria chosen: **Multiple Base Choice Coverage (MBCC)**

Base Case test cases: **A2, B2, C2, D3 & A2, B3, C3**

Tests	Blocks Covered	Status
T1	A2, B2, C2, D3	Pass
T2	A1, C2, D3	Pass
T3	A2, B1, C2, D3	Pass
T4	A2,B3, C2, D3	Pass
T5	A2, B2, C1	Pass
T6	A2, B2, C2,D1	Pass
T7	A2, B2, C2,D2	Pass

3. STRING FORMATRIGHT (STRING S, STRING MASK)

Input Domain Modeling

Characteristics	Blocks		
A: String. Null	True	False	
B: String. Length	Empty	Equal to 1	Greater than 1
C: Mask. Null	True	False	
D: Mask. Length	Empty	Equal to 1	Greater than 1

Constraints: If the string or the mask is null -> length characteristics are not applicable.

Test Cases

The test coverage criteria chosen: **Multiple Base Choice Coverage (MBCC)**

Base Case test cases: **A2, B2, C2, D3 & A2, B3, C3**

Tests	Blocks Covered	Status
T1	A2, B2, C2, D3	Pass
T2	A1, C2, D3	Pass
T3	A2, B1, C2, D3	Pass
T4	A2,B3, C2, D3	Pass
T5	A2, B2, C1	Pass
T6	A2, B2, C2,D1	Pass
T7	A2, B2, C2,D2	Pass

A set of unsorted elements with multiple occurrences.

1. PUBLIC INT ADD(T ENTRY)

Input Domain Modeling

Characteristics	Blocks	
A: Entry is Null	True	False
B: Entry exists	True	False

Constraints:

- All Characteristics are compatible with each other.

Test Cases

The test coverage criteria chosen: **All Combinations Coverage**

Tests	Blocks Covered	Status
T1	A2,B2	Pass
T2	A2, B1	Pass
T3	A1, B2	Pass
T4	A2, B2	Pass

2. PUBLIC INT ADD(T ENTRY, INT MULT)

Input Domain Modeling

Characteristics	Blocks	
A: Entry is Null	True	False
B: Entry exists	True	False
C: Multiplicity of key	Less than or equal 0	Greater than 0

Constraints:

Test Cases

The test coverage criteria chosen: **All Combinations Coverage**

Tests	Blocks Covered	Status
T1	A2, B2, C2 *	Pass
T2	A2, B2, C1	Pass
T3	A2, B1, C2	Pass
T4	A2, B1, C1	Pass
T5	A1, B2, C2*	Pass
T6	A1, B2, C1	Pass
T7	A1, B1, C2	Pass
T8	A1, B1, C1	Pass

3. PUBLIC VOID ADDALL(MULTISET<T> OTHER)

Input Domain Modeling

Characteristics	Blocks		
A: other is Null	True	False	
B: Other is empty	True	False	
C: other overlaps	Partially overlaps	Completely overlaps	Disjoint

Constraints:

- If A1 (other is Null)=> B and C don't apply
- If B1 (other is empty) => C doesn't apply

Test Cases

The test coverage criteria chosen: **Base Choice Coverage (BCC)**

Tests	Blocks Covered	Status
T1	A1	Pass
T2	A2, B1	Pass
T3	A2, B2, C1	Pass
T4	A2, B2, C2	Pass
T5	A2, B2, C3	Pass

4. PUBLIC VOID ADDALL(COLLECTION<T> VALUES)

Input Domain Modeling

Characteristics	Blocks		
A: other is Null	True	False	
B: other is empty	True	False	
C: other overlaps	Partially overlaps	Completely overlaps	Disjoint

Constraints:

- If A1 (other is Null)=> B, C don't apply
- If B1 (other is empty) => C don't apply

Test Cases

The test coverage criteria chosen: **Base Choice Coverage (BCC)**

Tests	Blocks Covered	Status
T1	A1	Pass
T2	A2, B1	Pass
T3	A2, B2, C1	Pass
T4	A2, B2, C2	Pass
T5	A2, B2, C3	Pass

5. PUBLIC VOID ADDALL(COLLECTION<T> VALUES)

Input Domain Modeling

Characteristics	Blocks		
A: other is Null	True	False	

B: other is empty	True	False	
C: other overlaps	Partially overlaps	Completely overlaps	Disjoint

Constraints:

- If A1 (other is Null)=> B, C don't apply
- If B1 (other is empty) => C don't apply

Test Cases

The test coverage criteria chosen: **Base Choice Coverage (BCC)**

Tests	Blocks Covered	Status
T1	A1	Pass
T2	A2, B1	Pass
T3	A2, B2, C1	Pass
T4	A2, B2, C2	Pass
T5	A2, B2, C3	Pass

6. INTEGER REMOVE(OBJECT KEY)

Input Domain Modeling

Characteristics	Blocks		
A: Key. Null	True	False	
B: Key. Match type T	True	False	
C: Multiplicity of key	0 (Empty)	Equal to 1	Greater than 1

Constraints:

- If **A1** (key is null) -> then **B** is not applicable. The multiplicity for null key is handled separately (the map may contain a null key).
- If **A2** (key non-null) and **B2** (type does not match T) -> then **C** is not applicable.

Test Cases

The test coverage criteria chosen: **All Combinations Coverage (ACoC)**

Excluding invalid combinations

Tests	Blocks Covered	Status
T1	A1	Pass
T2	A2, B1, C1	Pass
T3	A2, B1, C2	Pass
T4	A2, B1, C3	Pass
T5	A2, B2	Pass

7. INT REMOVE(T KEY, INT MULT)

Input Domain Modeling

Characteristics	Blocks		
A: Key. Null	True	False	
B: Current Multiplicity of key (old)	0	Equal to 1	Greater than 1
C: mult Value	<=0	Equal to 1	Greater than 1

Constraints:

- If **A1**, **B1** only is applicable
- If **B1**, **C** can be anything (result is always 0).
- If **B1**, then **C2** removes it, **C2** does nothing, **C3** removes it completely (behaves the same as C=1).
- If **B3**, the behavior differs for **C1** (no change), **C2** (decrease by 1), **C3** (decrease by mult, possibly to 0 or positive).

Test Cases

The test coverage criteria chosen: **Multiple Base Choice Coverage (MBCC)**

Base test cases: A2, B2, C2 & A2, B3, C3

Tests	Blocks Covered	Status
T1	A2, B2, C2	Pass
T2	A2, B1, C2	Pass
T3	A2, B3, C2	Pass
T4	A2, B2, C1	Fail
T5	A2, B2, C3	Pass
T6	A1, B1, C2	Pass
T7	A2, B3, C3	Pass
T8	A2, B1, C3	Pass
T9	A2, B3, C1	Fail
T10	A1, B1, C3	Pass

T4 and T9 Fails because of mult < 0 makes the multiplicity of the key increase

For example, parameter mult = -1 and the current key have mult 1 so it will be $1 - -1 = 2$ which is wrong.

8. INT REMOVEALL(T KEY)

Input Domain Modeling

Characteristics	Blocks		
A: Key. Null	True	False	
B: Multiplicity of key	0	Equal to 1	Greater than 1

Test Cases

The test coverage criteria chosen: **All Combinations Coverage (ACoC)**

Tests	Blocks Covered	Status
T1	A1, C1	Pass
T2	A1, C2	Pass
T3	A1, C3	Pass
T4	A2, C1	Pass
T5	A2, C2	Pass
T6	A2, C3	Pass

9. MULTISSET<T> INTERSECT(MULTISSET<T> OTHER)

Input Domain Modeling

Characteristics	Blocks		
A: Other Null	True	False	
B: Other Size	0	Equal to 1	Greater than 1
C: Current Size	0	Equal to 1	Greater than 1
D: Intersection Exists	True	False	

Constraints:

- D1 is not applicable with A1, B1, C1
- A1 is only applicable with B1

Test Cases

The test coverage criteria chosen: **Pair-Wise Coverage (PWC)**

Tests	Blocks Covered	Status
T1	A1, B1, C1, D2	Pass
T2	A2, B1, C2, D2	Pass
T3	A1, B1, C3, D2	Pass
T4	A2, B2, C1, D2	Pass
T5	A1, B2, C2, D1	Pass
T6	A2, B2, C3, D1	Pass
T7	A2, B3, C1, D2	Pass
T8	A2, B3, C2, D1	Pass
T9	A1, B3, C3, D2	Pass

10. PUBLIC INT TOTALCOUNT()

Input Domain Modeling

Characteristics	Blocks		
A: Multiset. No of distinct elements	Empty	1	Greater than 1
B: Multiset. Multiplicity	All = 1	Some > 1	All > 1

Constraints:

- If length is empty then the multiplicity can't be greater than 1

Test Cases

The test coverage criteria chosen: **Multiple Base Choice Coverage (MBCC)**

Base test cases: A3, B1& A2, B2

Tests	Blocks Covered	Status
T1	A3, B1	Pass
T2	A1, B1	Pass
T3	A2, B1	Pass
T4	A3, B2	Pass
T5	A3, B3	Pass
T6	A2, B2	Pass

11. PUBLIC LIST<T>TOLIST()

Input Domain Modeling

Characteristics	Blocks		
A: Multiset. No of distinct elements	Empty	1	Greater than 1
B: Multiset. Multiplicity	All = 1	Some > 1	All > 1

Constraints:

- If length is empty then the multiplicity can't be greater than 1

Test Cases

The test coverage criteria chosen: **Base Choice Coverage (BCC)**

Base test cases: **A3, B1**

Tests	Blocks Covered	Status
T1	A3, B1	Pass
T2	A1, B1	Pass
T3	A2, B1	Pass
T4	A3, B2	Pass
T5	A3, B3	Pass

12. PUBLIC STRING TOSTRING()

Input Domain Modeling

Characteristics	Blocks		
A: Multiset. No of distinct elements	Empty	1	Greater than 1
B: Multiset. Multiplicity	All = 1	Some > 1	All > 1

Constraints:

- If length is empty then the multiplicity can't be greater than 1

Test Cases

The test coverage criteria chosen: **Base Choice Coverage (BCC)**

Base test cases: **A3, B1**

Tests	Blocks Covered	Status
T1	A3, B1	Pass
T2	A1, B1	Pass
T3	A2, B1	Pass
T4	A3, B2	Pass
T5	A3, B3	Pass
T6	A2, B2	Pass

13. PUBLIC BOOLEAN EQUALS(OBJECT O)

Input Domain Modeling

Characteristics	Blocks		
A: O. Null	True	False	
B: O. Type	Not a Multiset	Multiset	
C: Size Comparison	Different	Same	
D: Content Comparison	Same Keys, Multiplicities	Same Keys, Different Multiplicities	Different Keys

Constraints:

- **A1 blocks everything**
 - $A1 \rightarrow B, C, D$ are all infeasible
 - If o is null, we return false immediately, nothing else matters
- **Constraint 2: B1 blocks C and D**
 - $A2 + B1 \rightarrow C, D$ are infeasible
 - If o is not a Multiset, we return false immediately
- **Constraint 3: C1 blocks D**
 - $A2 + B2 + C1 \rightarrow D$ is infeasible
 - If sizes differ, we return false immediately before checking content

Test Cases

The test coverage criteria chosen: **All Combinations (ACoC)**

Tests	Blocks Covered	Status
T1	A1	Pass
T2	A2 , B1	Pass
T3	A2,B2,C1	Pass
T4	A2,B2,C2,D1	Pass
T5	A2,B2,C2,D2	Pass
T6	A2,B2,C2,D3	Pass